Aug 31 2019

Final Exam

45	24	21	0	53.33%	60:00	24
Questions	Right	Wrong	Unattended	Accuracy	Time Taken	Marks Scored

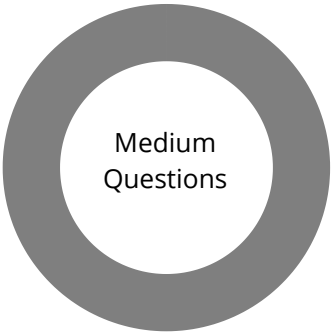
View rank list (https://elearn.maven-silicon.com/report/assessment/103)

Topics	No.of Qns	Right	Wrong	Unattended	Accuracy	Remarks
Design Flow	5	3	2	0	60%	Needs Improvement
Digital Electronics	20	10	10	0	50%	Offline
Verilog	20	11	9	0	55%	Needs Improvement

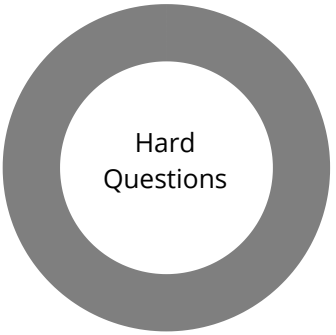
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Right Answers	24
Wrong Answers	21
Unattended	0



Right Answers	0
Wrong Answers	0
Unattended	0



Right Answers	0
Wrong Answers	0
Unattended	0

All questions

Compare (https://elearn.maven-silicon.com/report/assessment/103)

Showing **45** of **45** questions

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1 Single choice

The design flow in VLSI Systems is in which order?

1. Architecture Design

2. Market requirement

3. Verification

4. HDL coding

a) **2 – 1 – 4 – 3**

b) **4 – 1 – 3 – 2**

c) **3 – 2 – 1 – 4**

d) **1 – 2 – 3 – 4**

e) **3 – 1 – 2 – 4**

Your answer (a) **Right**

Time Taken **30s** **Marks 1** **Answer Explanation**

2 Single choice

Gate level Simulation is to ensure that, _____

a) **Both RTL and Gate level netlist have same no. of input & output lines**

b) **the synthesis tool has no bugs in it's implementation**

c) **Both RTL & Gate level netlist are equivalent in terms of functionality**

d) **All of the above.**

e) **None of the above.**

Your answer (C) **Right** Time Taken **15s** **Marks 1** **Answer Explanation**

3 Single choice

The structures like H – tree and I- tree are used for _____

a) **Increasing the clock slew rate**
b) **Increasing the speed of the circuit**
c) **Minimizing the clock skew effect**
d) **All of the above.**
e) **None of the above.**

Your answer (b) **Wrong** Right answer (c) Time Taken **17s** **Marks 0** **Answer Explanation**

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4 Single choice

The Layout Vs Schematic (LVS) is done to ensure that, _____

a) **The functionality of schematic and Layout are equivalent**
b) **Layout occupies lesser silicon area**
c) **Layout is drawn by considering all t he design rules.**
d) **All of the above**
e) **None of the above**

Your answer (c) **Wrong** Right answer (a) Time Taken **57s** **Marks 0** **Answer Explanation**

5 Single choice

While defining specification for designing ASIC's, we need to include _____

a) **Input & Output specification**
b) **Power requirements**
c) **Speed requirements**
d) **All of the above**
e) **None of the above**

Your answer (d) **Right** Time Taken **14s** **Marks 1** **Answer Explanation**

6 Single choice

Find the radix (512/41)r = (12) r

a) **0**
b) **6**
c) **8**
d) **9**
e) **none of these**

Your answer (C) **Right** Time Taken **115s** **Marks 1** **Answer Explanation**

7 Single choice

$(10000)_{16} - (1001)_{16}$ is

a) **EFFF**
b) **00111**
c) **both a and b**
d) **01000**
e) **none of these**

Your answer (a) **Right** Time Taken **129s** **Marks 1** **Answer Explanation**

8 Single choice

Which of the expression are equivalent to A^B^C^D?

i. ***F(A,B,C,D)=∑(1,2,4,7,8,11,13,14)***

ii. ***F(A,B,C,D)=∑(3,5,7,10,11,12,13,14)***

iii. ***F(A,B,C,D)=π(0,3,5,6,9,10,12,15)***

iv. ***F(A,B,C,D)=π(0,1,2,4,6,8,9,15)***

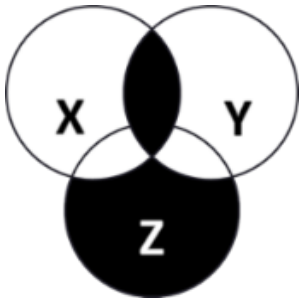
a) **ii and iii**
b) **iv and i**
c) **ii and iv**
d) **i and iii**
e) **None of the above**

Your answer (d) **Right** Time Taken **106s** **Marks 1** **Answer Explanation**

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9 Single choice

The Boolean expression for shaded area in Venn Diagram is _____



a) **X'+Y'+Z**

b) **XY'Z+X'YZ**

c) **X+Y+Z**

d) **X'Y'Z+XY**

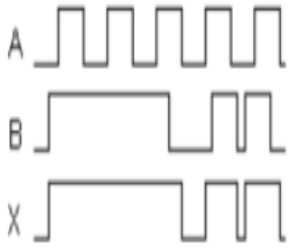
e) **None of the above**

Your answer (b) **Wrong** Right answer (d) Time Taken **239s** **Marks 0** **Answer Explanation**

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10 Single choice

Design the circuit to get X output with the help of the following waveform. A and B are Inputs of _____?



a) **OR Gate**

b) **AND Gate**

c) **XOR Gate**

d) **XNOR Gate**

e) **None of the above**

Your answer (a) **Right** Time Taken **54s** **Marks 1** **Answer Explanation**

11 Single choice

A 32 bit ripple carry adder is realized using 32 identical full adders (FA). The carry propagation delay of each FA is 10 ns and the sum propagation delay of each FA is 15 ns. Worst case delay of this 32 bit adder is _____

a) **325ns**

b) **480ns**

c) **Both can be right**

d) **400ns**

e) **None of the above**

Your answer (b) **Wrong** Right answer (a) Time Taken **76s** **Marks 0** **Answer Explanation**

12 Single choice

A priority encoder is an encoder circuit with priority. If 2 or more inputs are equal to 1, the input with the highest priority will take precedence. The truth table of a 4-input priority encoder is shown below. An output V is added which is set to1 when one or more inputs are equal to 1; otherwise V is 0. The two outputs x and y are not inspected when V equals 0 and hence they are specified as don′t care outputs. Note that whereas x′s in output columns represent don′t care outputs, the x′s in the input columns are useful for representing a truth table in compact form. For example, input X100 represents the 2 input combinations: 0100 and 1100. The simplified SOP expressions for Y is_____.

Inputs				Outputs		
D3	D2	D1	D0	X	Y	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

a) **D0+D2D1'**

b) **D1D2+D2D4'**

c) **D0+D4D3'**

d) **D1+D2D3'**

e) **None of the above**

Your answer (b) **Wrong** Right answer (a) Time Taken **145s** **Marks 0** **Answer Explanation**

13 Single choice

How many 4 to 1 multiplexers are required to implement 100 to 1 multiplexer?

a) **33**

b) **34**

c) **32**

d) **36**

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e) **None of the above**

Your answer (b) **Wrong** Right answer (a) Time Taken **45s** **Marks 0** **Answer Explanation**

14 Single choice

How many minimum nor gates are required to implement function?

$$f(A,B,C,D) = \sum m(1,2,4,5,6,10,12,14)$$

a) **6**

b) **7**

c) **8**

d) **9**

e) **None of the above**

Your answer (a) **Wrong** Right answer (b) Time Taken **136s** **Marks 0** **Answer Explanation**

15 Single choice

Product-of-Sums expressions can be implemented using _____

a) **2-level OR-AND logic circuits**

b) **2-level NOR logic circuits**

c) **2-level XOR logic circuits**

d) **Both 2-level OR-AND and NOR log ic circuits**

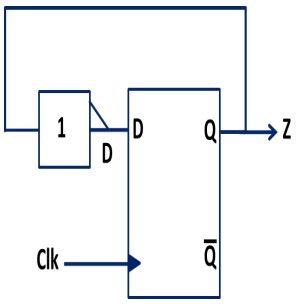
e) **None of the above**

Your answer (d) **Right** Time Taken **122s** **Marks 1** **Answer Explanation**

16 Single choice

The circuit below forms a ÷2 counter. If the inverter has a propagation delay of 5 ns and the propagation delay, setup time and hold time of the flipflop are 8 ns, 4 ns and 2 ns respectively, calculate the highest clock frequency for reliable operation.

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a) **58.8MHz**

b) **68.8MHz**

c) **49.8MHz**

d) **53.8MHz**

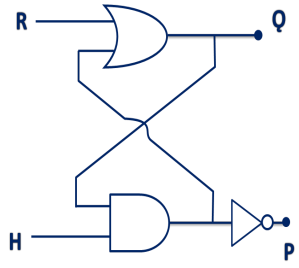
e) **None of the above**

Your answer (a) **Right** Time Taken **169s** **Marks 1** **Answer Explanation**

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17 Single choice

A latch can be constructed from an OR gate, an AND gate, and an inverter connected as follows:



What restriction must be placed on R and H so that P will always equal Q′ (under steady-state conditions)?

a) **00**

b) **01**

c) **10**

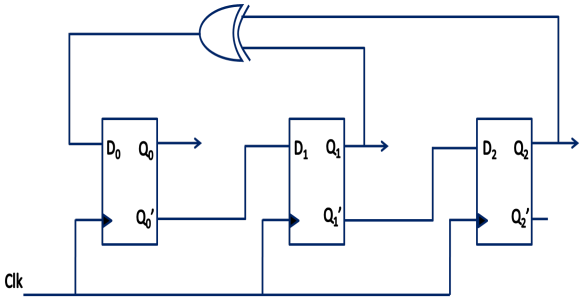
d) **11**

e) **None of the above**

Your answer (C) **Right** Time Taken **135s** **Marks 1** **Answer Explanation**

18 Single choice

Consider the synchronous sequential circuit shown below. Which states are not reachable from initial state S4 (Q0, Q1, Q2) = 100? What is the modulus of the counter?



a) **5**

b) **6**

c) **7**

d) **4**

e) **None of the above**

Your answer (a) **Wrong** Right answer (c) Time Taken **157s** **Marks 0** **Answer Explanation**

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19 Single choice

How many address bits are required for a 4096-bit memory organized as a 512 × 8, bit addressable memory?

a) **12 address lines**

b) **2^9 address lines**

c) **9 address lines**

d) **2^12 address lines**

e) **None of the above**

Your answer (a) **Right** Time Taken **40s** **Marks 1** **Answer Explanation**

20 Single choice

Assume that a 4-bit serial in/serial out left shift register is initially clear. We wish to store the nibble 1100. What will be the output pattern after the fourth clock pulse? (Left-most bit first)?

a) **0000**

b) **0011**

c) **1100**

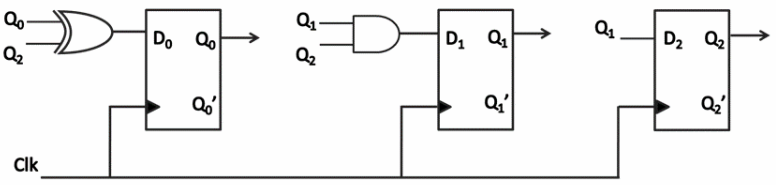
d) **1001**

e) **None of the above**

Your answer (a) **Wrong** Right answer (b) Time Taken **52s** **Marks 0** **Answer Explanation**

21 Single choice

Consider the synchronous sequential circuit shown below.



Which states are not reachable from initial state S5(Q0, Q1, Q2) = 101?

a) **000,011,100 and 111are not reachable**

b) **000,010,100 and 111are not reachable**

c) **000,011,110 and 111are not reachable**

d) **000,011,101 and 111are not reachable**

e) **None of the above**

Your answer (C) **Wrong** Right answer (e) Time Taken **245s** **Marks 0** **Answer Explanation**

22 Single choice

How many address bits are required for a 4096 KB memory which has a data width capacity of 16 bits?

a) **21 address lines**

b) **12 address lines**

c) **10 address lines**

d) **18 address lines**

e) **None of the above**

Your answer (C) **Wrong** Right answer (a) Time Taken **149s** **Marks 0** **Answer Explanation**

23 Single choice

What is the modulus of the 6-bit ring counter ?

a) **2**

b) **4**

c) **6**

d) **12**

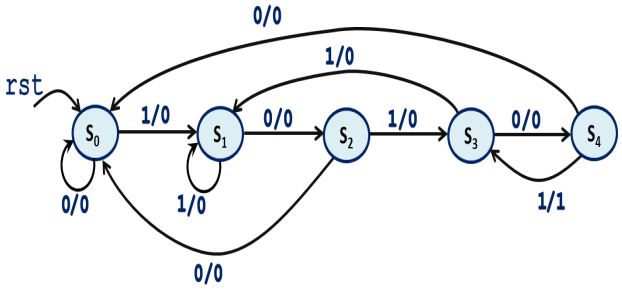
e) **None of the above**

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Your answer (C) **Right** Time Taken **9s** **Marks 1** **Answer Explanation**

24 Single choice

Which of the following is right with respect to the given FSM?



a) **to detect overlapping sequence "101 01"**

b) **to detect non overlapping sequence "10101"**

c) **to detect non overlapping sequence "1010"**

d) **to detect non overlapping sequence "1011"**

e) **none of the above**

Your answer (a) **Right** Time Taken **152s** **Marks 1** **Answer Explanation**

25 Single choice

What is the minimum number of states required to design a Moore FSM with input as a bit stream, and o/p which goes 1 whenever the number received so far is divisible by 3?

a) **5**

b) **8**

c) **3**

d) **9**

e) **None of the above**

Your answer (a) **Wrong** Right answer (c) Time Taken **148s** **Marks 0** **Answer Explanation**

26 Single choice

Which abstraction level is used by the most of the synthesizer tools?

a) **Dataflow level**

b) **Gate level**

c) **Behavioral level**

d) **RTL level**

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e) **None of the above**

Your answer (d) **Right** Time Taken **23s** **Marks 1** **Answer Explanation**

27 Single choice

What is an error in the following name based port mapping?

```
half_adder_tb DUT(.a(a),,.sum(sum),.cout(cout)) ;
```

a) **One extra comma is given**

b) **There is no error**

c) **Port mapping should be order based**

d) **During port mapping none of the ports should be kept floating**

e) **None of the above**

Your answer (a) **Wrong** Right answer (d) Time Taken **32s** **Marks 0** **Answer Explanation**

28 Single choice

What value is stored in the variable "y" after 10ns?

```
integer y;
initial
  begin
    y = -2 ;
    #10;
    y = -'d12/2;
  end
```

a) **-2**

b) **-6**

c) **2's complement of -6**

d) **2147483642**

e) **None of the above**

Your answer (b) **Wrong** Right answer (d) Time Taken **27s** **Marks 0** **Answer Explanation**

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29 Single choice

What is the value passed to the port "ain" from the below snippet?

```
module driver(output [63:0]ain);
  real r ;

  assign ain = r ;
  initial
    begin
        #10;
        r = 4.3 ;
    end
endmodule
```

a) **4.3**

b) **4**

c) **5**

d) **0**

e) **None of the above**

Your answer (b) **Right** Time Taken **54s** **Marks 1** **Answer Explanation**

30 Single choice

What is the syntax to access the 4th bit from LSB of an array with size 10x8?

```
reg [7:0] array [9:0];
```

a) **array[4]**

b) **array[addr][3]**

c) **array[addr][4]**

d) **array[addr][8]**

e) **None of the above**

Your answer (d) **Wrong** Right answer (b) Time Taken **44s** **Marks 0** **Answer Explanation**

31 Single choice

Which is true from the following statements for Parameters?

a) **Parameters are constants which can change anytime.**

b) **Parameters are variables**

c) **Parameters are global constants.**

d) **All of the above**

e) **None of the above**

Your answer (a) **Wrong** Right answer (c) Time Taken **18s** **Marks 0** **Answer Explanation**

32 Single choice

What is the output of the following snippet?

```
reg y;
initial
  begin
    y = 1'bx && 2'b11;
  end
```

a) **y = 1**

b) **y = 0**

c) **y = x**

d) **y = z**

e) **None of the above**

Your answer (C) **Right** Time Taken **12s** **Marks 1** **Answer Explanation**

33 Single choice

What is the output of the following snippet?

```
reg [2:0]y;
initial
  begin
    y = ~&4'b1101;
  end
```

a) **y = 3'b001**

b) **y = 3'b000**

c) **y = 3'b100**

d) **y = 3'b111**

e) **None of the above**

Your answer (d) **Wrong** Right answer (a) Time Taken **30s** **Marks 0** **Answer Explanation**

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34 Single choice

What is the value of a from the following snippet?

```
reg [3:0]a;
initial
  begin
    a = {2'bx,1'b1};
  end
```

a) **a = 4'b00x1**

b) **a = 4'b0xx1**

c) **a = 4'bxxx1**

d) **a = 4'bx0x1**

e) **None of the above**

Your answer (b) **Right** Time Taken **71s** **Marks 1** **Answer Explanation**

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35 Single choice

What is the value of a from the following snippet?

```
wire [3:0]a;
assign  a = (2'b0x == 2'b00) ? 4'b1100:4'b1101;
```

a) **a = 4'b1101**

b) **a = 4'b110x**

c) **a = 4'b1100**

d) **a = 4'b1111**

e) **None of the above**

Your answer (e) **Wrong** Right answer (b) Time Taken **92s** **Marks 0** **Answer Explanation**

36 Single choice

Which is not true from the following statements?

a) **Continuous assignments are concurrent**

b) **Continuous assignments are made to Net type only**

c) **The RHS of a continuous assignments can be reg or wire.**

d) **All of the above**

e) **None of above**

Your answer (a) **Wrong** Right answer (e) Time Taken **24s** **Marks 0** **Answer Explanation**

37 Single choice

Which is true from the following statements?

a) **Procedural assignments are made to Integers only.**

b) **Initial block is non-synthesizable.**

c) **Always block will run once.**

d) **All of above**

e) **None of above**

Your answer (e) **Wrong** Right answer (b) Time Taken **63s** **Marks 0** **Answer Explanation**

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38 Single choice

At what time value of a is updated to 6 from the below snippet?

```
reg [3:0]a;
initial
  begin
    a = 4'd4;
    #10;
    a = 4'd6;
    #20;
  end
```

a) **20ns**

b) **10ns**

c) **0ns**

d) **30ns**

e) **None of above**

Your answer (b) **Right** Time Taken **27s** **Marks 1** **Answer Explanation**

39 Single choice

At what time value of a is updated to 8 from the below snippet?

```
reg [3:0]a;
initial
  begin
    a = #10 4'd4;
    a = #20 4'd8;
    #20;
  end
```

a) **20ns**

b) **10ns**

c) **30ns**

d) **50ns**

e) **None of above**

Your answer (C) **Right** Time Taken **45s** **Marks 1** **Answer Explanation**

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40 Single choice

At what time value of a is updated to 6 from the below snippet?

```
reg [3:0]a;
initial
  begin
    a <= #10 4'd4;
    a <= #20 4'd6;
    #20;
  end
```

a) **20ns**

b) **10ns**

c) **30ns**

d) **50ns**

e) **None of above**

Your answer (a) **Right** Time Taken **23s** **Marks 1** **Answer Explanation**

41 Single choice

At what time value of a is updated to 8 from the below snippet?

```
reg [3:0]a;
initial
 begin
    a <= 4'd4;
    #10;
    a <= 4'd8;
    #20;
 end
```

a) **20ns**

b) **10ns**

c) **0ns**

d) **30ns**

e) **None of the above**

Your answer (b) **Right** Time Taken **69s** **Marks 1** **Answer Explanation**

42 Single choice

What type of case the below snippet belongs to?

```
reg[1:0]a;
initial
    begin
      case(a)
       2'b00 : st1;
       2'b01 : st2;
       2'b10 : st3;
      endcase
    end
```

a) **Full case**

b) **Parallel case**

c) **Overlapping case**

d) **Non overlapping case**

e) **None of the above**

Your answer (b) **Right** Time Taken **41s** **Marks 1** **Answer Explanation**

43 Single choice

For inferring a priority encoder logic, which construct should be used?

a) **if-else**

b) **case**

c) **for loop**

d) **while loop**

e) **None of the above**

Your answer (a) **Right** Time Taken **53s** **Marks 1** **Answer Explanation**

44 Single choice

For inferring a single Mux logic, which construct should be used?

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a) **if-else**

b) **case**

c) **for loop**

d) **while loop**

e) **None of the above**

Your answer (b) **Right**

Time Taken **44s** **Marks 1** **Answer Explanation**

45 Single choice

In a Moore based FSM, which of the following statement is true?

a) **Outputs should be reg type.**

b) **Outputs shouldn't depend on clock edge .**

c) **Glitches can occur in the output.**

d) **All of the above**

e) **None of the above.**

Your answer (a) **Wrong** Right answer (b)

Time Taken **64s** **Marks 0** **Answer Explanation**

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